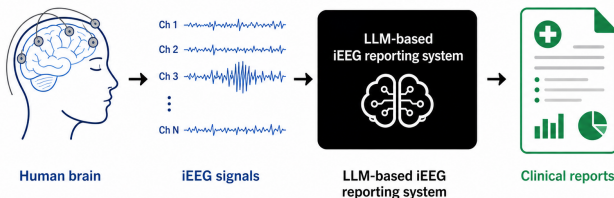


Automatic Generation of Text-Based Summaries of Intracranial EEG Seizure Events Using LLMs

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Background

- ▶ **Epilepsy** is a brain-network disorder characterized by recurrent seizures [1].
- ▶ Almost 30% of epilepsy patients are **pharmacoresistant** [2].
- ▶ **Epilepsy surgery requires precise localization** and its success depends on identifying the epileptogenic zone/seizure onset zone (SOZ).
- ▶ **HFOs** are brief high-frequency events that may mark epileptogenic tissue.
- ▶ **Seizure detection** identifies when ictal activity occurs in the recording.
- ▶ **iEEG has high spatial and temporal resolution** that supports analysis of HFOs and seizure events [3].

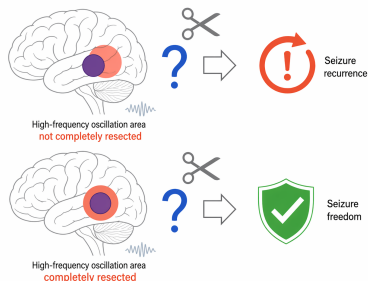


Figure 1: Resection of HFO-related area and seizure outcome.

Motivation: iEEG interpretation is clinically important, but manual analysis is time-consuming and difficult to standardize. This motivates automated analysis of HFO and seizure evidence and structured report generation.

Literature review: what prior work motivates

1. HFO biomarkers

Prior work:

HFOs: 80–500 Hz.

Ripples: 80–250 Hz.

Fast ripples: 250–500 Hz.

FRandR is more specific than ripples alone [4].

Limitation:

Physiological HFOs can mimic pathological HFOs (e.g., sensory-motor or cognitive activity).

Design choice:

Use ripple, FR, and FRandR channel rates.

2. Seizure detection

Prior work:

Seizure detection classifies ictal (during seizure) and non-ictal (outside seizure) windows.

CNNs capture local patterns; RNN/LSTM models capture temporal changes; transformers model longer context.

Limitation:

Subject variability and class imbalance remain difficult.

Design choice:

Use a CNN-transformer seizure detector.

3. LLM reporting

Prior work:

Recent EEG-agent and AI-reporting systems show that LLMs can support tool use, structured reasoning, and report generation from extracted EEG features.

Limitation:

Most systems focus on scalp EEG or general EEG workflows [5, 6] .

Design choice:

Use a JSON-grounded Qwen3:8B model pipeline for reasoning and structured iEEG report generation.

Research gap, question, and significance

Gap in prior work

- ▶ HFO detection, seizure detection, and reporting are usually treated **separately**.
- ▶ LLM-based EEG reporting exists mainly for **scalp EEG**, not invasive iEEG.
- ▶ There is **no universally standardized iEEG reporting template**.

Research question

Can a unified pipeline combine HFO detection, seizure detection, and LLM-based reasoning to produce structured, evidence-grounded iEEG reports?

Significance

A successful system could improve standardization, reduce reporting burden, and support pre/post-surgical interpretation.

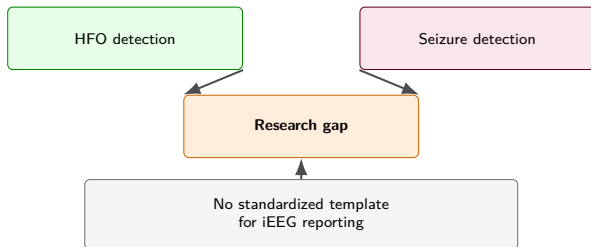


Figure 2: Research gap.

Proposed approach

Core idea: A unified framework combines HFO detection, seizure detection, and an LLM-based reporting system to generate structured iEEG reports.

1. Evidence reasoning

Interprets HFO and seizure findings together.

2. Uncertainty checking

Checks mismatch, uncertainty, and evidence consistency.

3. Structured reporting

Generates the final evidence-grounded iEEG report.

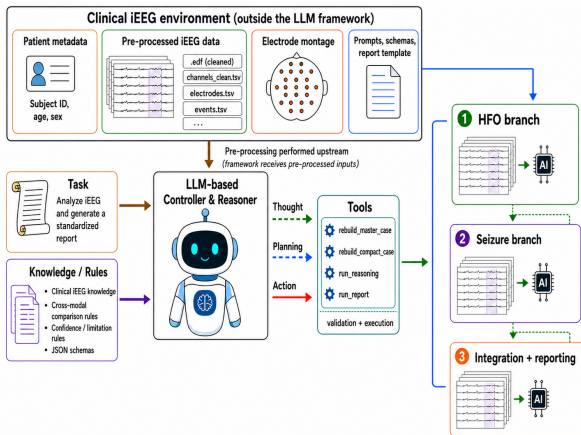


Figure 3: General LLM system framework.

Methods

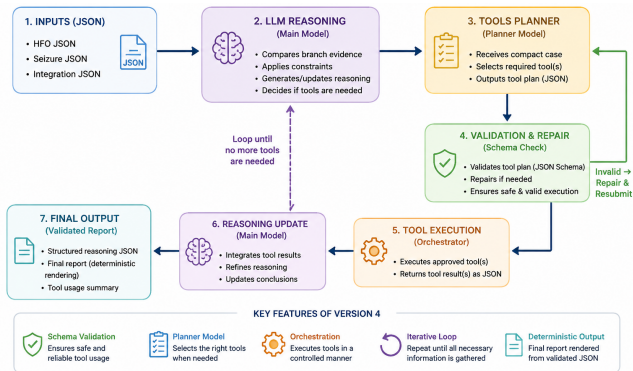


Figure 4: Pipeline overview.

Version 4 uses bounded tools: *rebuild master case*, *rebuild compact case*, *run reasoning*, and *regenerate report*.

Datasets used

- ▶ **OpenNeuro ds003498 [8]**: initial HFO development.
- ▶ **SWEC-ETHZ**: initial seizure-model exploration.
- ▶ **HUP iEEG dataset, OpenNeuro ds004100 [7]**: final integration dataset with 57 presurgical drug-resistant focal epilepsy patients, variable sampling rates (256, 500, 512, 1024 Hz), and expert seizure onset/offset annotations.

HFO detection branch

HFO branch goal: identify interictal channel-level biomarkers that may support epileptogenic-zone localization.

- ▶ **Input:** preprocessed interictal iEEG recordings (made with Notch filter, Common Average Referencing, i.e., CAR)
- ▶ **Filtering:** ripple band and FR band analysis.
- ▶ **Detection:** candidate ripple/FR events with artifact rejection and stricter HUP constraints.
- ▶ **Pairing:** temporally overlapping ripple + FR events treated as FRandR.
- ▶ **Output:** per-channel rates, 95th-percentile HFO-area candidates, JSON summary.

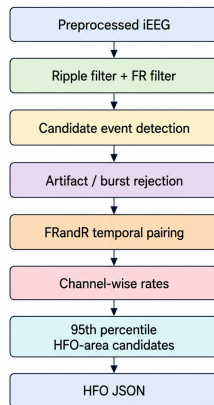


Figure 5: HFO detection architecture.

Seizure detection branch

Seizure branch goal: detect ictal windows and export structured seizure evidence for LLM integration.

- ▶ **Input:** preprocessed HUP iEEG windows labelled as ictal or non-ictal.
- ▶ **Windowing:** final model used 3-s windows after resampling.
- ▶ **CNN encoder:** extracts local waveform features from each channel.
- ▶ **Transformer:** models relationships between channel-level features.
- ▶ **Decision:** sigmoid probability, smoothing, and tuned threshold.
- ▶ **Output:** ictal segments, probabilities, metrics, saliency proxy, and seizure JSON.

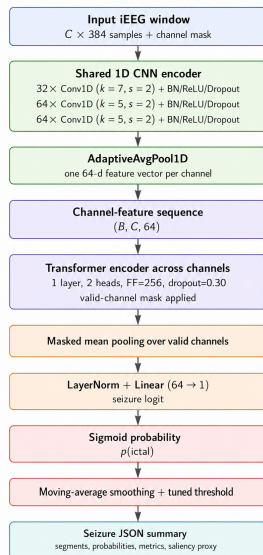


Figure 6: Seizure detection architecture.

Evolution of the LLM reporting system

Goal

Make the LLM output more structured, faithful to JSON evidence, and reproducible.

Development of the LLM reporting pipeline

Ver.	Main role	Key improvement
V1	Report generation from JSON	First structured report generated from HFO and seizure detector outputs.
V2	Seizure-channel evidence + fusion	Improved comparison between HFO and seizure evidence, including agreement or mismatch.
V3	Schema validation + deterministic rendering	Reduced free-form LLM errors and made the final report more reproducible.
V4	Bounded tool orchestration	Added controlled planner-based execution for rebuilding, reasoning, validation, and report regeneration.

Takeaway: The LLM component evolved from simple report generation to validated, evidence-grounded reasoning with controlled orchestration.

Main HFO findings

- ▶ FRandR timing offsets clustered near 0 ms, supporting meaningful co-occurrence pairing.
- ▶ Higher-frequency biomarkers were more **focal** than ripples alone.
- ▶ Cross-run summaries showed stronger concentration for FR/FRandR than for ripple rates.
- ▶ The branch produced channel-level candidate HFO areas for later integration.

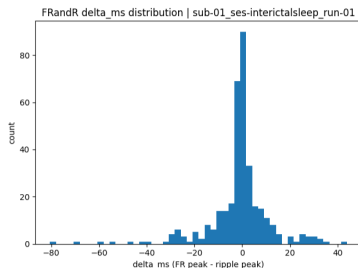


Figure 7: FRandR distribution.

HFO-area candidates selected by the 95th-percentile threshold

Band	Selection rule	Channel	Count	Rate/min	Threshold
FR	95p, 2/26 channels	PHR1	97	20.42	9.53
FR	95p, 2/26 channels	PHR2	50	10.53	9.53
FRandR	95p, 2/25 channels	PHR1	97	20.42	10.11
FRandR	95p, 2/25 channels	PHR2	51	10.74	10.11

Main seizure-detection findings

Cohort-level result

- ▶ Across 43 subjects, mean subject-level accuracy and balanced accuracy reached **0.83**; median values were **0.87**.
- ▶ Median F1-score was **0.84**; median AUROC and AUPRC were **0.97** and **0.95**.

Interpretation

- ▶ Performance was strong for many subjects, but heterogeneous across the cohort.
- ▶ Difficult or unstable cases show that generalization remains challenging.

Representative subject-level results

Group	Subject	Runs	Acc.	Bal. Acc.	F1	AUROC	Std F1
Strong	sub-HUP160	3	0.997	0.996	0.995	1.000	0.005
Strong	sub-HUP088	3	0.989	0.988	0.992	0.999	0.005
Strong	sub-HUP142	3	0.989	0.988	0.988	0.999	0.004
Moderate	sub-HUP074	5	0.972	0.974	0.968	0.997	0.023
Moderate	sub-HUP080	4	0.976	0.978	0.972	0.999	0.019
Moderate	sub-HUP106	3	0.956	0.958	0.956	0.980	0.019
Difficult / unstable	sub-HUP070	5	0.809	0.838	0.601	0.904	0.310
Difficult / unstable	sub-HUP117	3	0.652	0.543	0.277	0.450	0.440
Difficult / unstable	sub-HUP181	5	0.431	0.510	0.418	0.428	0.282

Case study: HUP146 and the final reporting pipeline

Why this case matters

- ▶ Main demonstration subject for the LLM pipeline.
- ▶ Final seizure-detection result on the held run was strong:
 - ▶ Accuracy: **0.963**
 - ▶ Balanced accuracy: **0.928**
 - ▶ F1-score: **0.922**

What the LLM stage adds

- ▶ Integrates branch summaries.
- ▶ Preserves uncertainty when modalities disagree.
- ▶ Produces a structured final report.

Version 4 final report for subject HUP146

Case overview. Subject sub-HUP146 was analyzed with integrated HFO and seizure branches. Cross-modal comparison used exact channel-name matching.

HFO findings. HFO consensus strength was strong. Strong consensus channels were LG8 and RC5. Additional supportive channels were RI11, RJ5, LG7, LG5, and LG4. A total of 246 FR events and 18 FRandR pairs were detected across 89 epochs.

Seizure findings. Two ictal segments were detected: 119–132.5 s and 136.5–199 s. Peak probabilities were 0.88 and 0.91. Seizure channel evidence was treated as a model-derived saliency proxy.

Cross-modal integration. Concordance level was none. No exact channel overlap was found between HFO channels and seizure-supportive channels, producing a mismatch warning.

Limitations and confidence. Overall confidence was moderate. Strong HFO consensus and high-probability ictal detections support confidence, but the cross-modal mismatch limits the final interpretation.

Final impression. Insufficient direct channel overlap prevents a definitive EZ hypothesis.

Figure 8: Excerpt from the Version 4 generated report for HUP146.

Final LLM evaluation results

Subject	Reasoning	Faithfulness	Orchestration
sub-HUP126	10/10	10/10	7/7
sub-HUP130	10/10	10/10	7/7
sub-HUP146	10/10	10/10	7/7
sub-HUP157	8/10	9/10	7/7
sub-HUP164	10/10	10/10	7/7
Average	9.6/10	9.8/10	7.0/7

Evaluation process

Note: this was a technical evaluation based on JSON/report consistency; no clinical expert review was performed.

1. Reasoning quality

Scored with a 0–2 rubric across evidence coverage, cross-modal reasoning, uncertainty handling, constraint adherence, and clarity.

2. Faithfulness

Checked whether channels, timings, confidence, and conclusions were supported by the compact JSON.

3. Orchestration validity

Checked whether the planner output was valid JSON, used allowed tools, followed order, and regenerated valid outputs.

Takeaway: The final system produced mostly high-quality, faithful, and valid outputs across the evaluated subjects.

Contributions and limitations

Contributions

- ▶ Developed a **unified pipeline** linking HFO analysis, seizure detection, and LLM-based reporting.
- ▶ Adapted HFO and seizure-detection pipelines to **HUP iEEG data**.
- ▶ Improved the LLM stage from plain generation to **validated, evidence-grounded reasoning**.
- ▶ Demonstrated a step toward **automated and standardized iEEG reporting**.

Limitations

- ▶ Performance remains **heterogeneous across subjects**.
- ▶ Seizure channel evidence is a **model-derived saliency proxy**, not definitive clinical onset.
- ▶ Cross-modal disagreement lowers confidence and should not be **over-interpreted**.
- ▶ The final report is technical support, not a clinical decision by itself.

Future work

- ▶ Improve robustness across difficult subjects.
- ▶ Strengthen channel-level seizure localization.
- ▶ Evaluate on more cases and in clinically realistic workflows.
- ▶ Refine standardized reporting templates for iEEG.

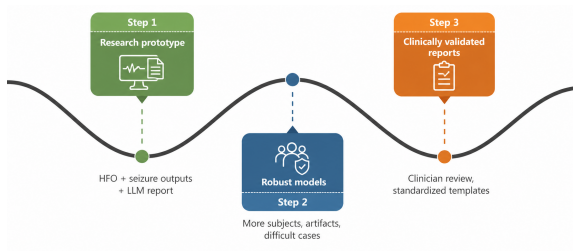


Figure 9: Roadmap from research prototype to clinically validated reporting.

Takeaway: The system is a research prototype; the next step is stronger robustness testing and clinical validation.

Conclusion

Problem

Manual iEEG interpretation is time-consuming, subjective, and difficult to standardize.

Answer

This thesis developed a unified HFO, seizure detection, and LLM-based reporting pipeline.

Impact

The final system produced evidence-grounded and faithful reports across evaluated subjects.

Final message: The proposed framework is a research prototype showing how iEEG analysis can move toward more structured, reproducible, and clinically usable reporting.

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Thank you

Questions?

